

Sections :H,I

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)

II Sem B.Tech. (CBCS) Mid Semester Examinations - I, February 2023

MA1003-1 – DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORMS

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Find the orthogonal trajectory of the family of curves $\frac{x^2}{a^2} + \frac{y^2}{a^2 + \lambda} = 1$, λ is a parameter.	5	L*2	1	1
b)	Solve $x(x - y)dy + y^2dx = 0$.	5	L2	1	1
2. a)	Solve $p(p + y) = x(x + y)$	5	L2	1	1
b)	If a body which is originally at $90^\circ C$ cools down to $70^\circ C$ in 20 minutes, the temperature of air being $40^\circ C$. Find the temperature of the body after 40 minutes from the original.	5	L2	1	1
Unit – II					
3. a)	Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 4\cos^2x$.	5	L1	2	1
b)	Solve $D(D + 1)^2y = 12e^{-x}$.	5	L2	2	1
4. a)	Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 4y = 2x^2 + 3e^{-x}$.	5	L1	2	1
b)	Solve $(D^2 + 4D + 4)y = 3\sin x + 4\cos x$	5	L2	2	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
